

TA brief

No cold calling; nobody is asked to read difficult text aloud.

Beside the You do, mark I (independent), S (supported) or C (choice): the level of independence, not the effort.

Give time to start → point to the relevant place → rephrase → offer a small prompt or word bank → reduce to a limited choice. Never straight to telling the answer.

Tell the teacher the barrier: reading, vocabulary, processing, regulation, writing or a science misconception. Avoid prompting through every question or dictating the final answer.

If behind

A pupil who missed last lesson: give the Arrival reminders and today's must-know words, not the missed lesson's work.

Photograph the practical at the planned point, upload it to the school's evidence app with this code, and stick a copy in the book.

Lesson code: GROW-SCI-A1-W08-P1. Planned photograph: after recording observations and before the conclusion.

P17 · SLT planning and evaluation

Lesson code : GROW-SCI-A1-W08-P1

SoW reference : GROW scheme of work, Autumn 1, return lesson

Exact SoW outcome : Describe the movement of the Moon around the Earth.

Evidence (scheme of work) : Pearson Edexcel Entry Level Certificate in Science 8939 (E1 → E3); AQA UAS science units. Working-scientifically observation, explanation and recorded evidence.

Evidence in this lesson : State that the Moon orbits Earth in roughly a month.; Explain that moonlight is reflected sunlight.; Explain phases using our changing view of the lit half.

Evidence location : Book work beneath the strip, retained first answer and pupil-owned edit. Keep spoken, practical or photographic evidence under the same code.

Label evidence on the school's evidence app with this code : GROW-SCI-A1-W08-P1

Arrival task (4 min) : Task: In your book: Glue in your strip; answer four prompts. Staff move: Read four prompts, sample privately and choose one route. IF NEW: use provided reminders. IF REVISION: retrieve before opening help.

Starter · choose a first idea (4 min) : Task: In your book: Keep one first idea and a reason. Staff move: Receive a first idea without upgrading the pupil's meaning; return to it after the evidence.

I do · make the science visible (4 min) : Task: In your book: Watch the model; name the evidence. No writing yet. Staff move: Keep the visible model and language linked. Pause at one observation; distinguish what the model shows from its limit.

We do · everyone answers (4 min) : Task: In your book: Choose an answer; explain the evidence clue. Staff move: Everyone commits. Secure: transfer. Mixed: compare two choices. Misconception: re-model, then use a fresh example. Access: change response mode.

I do · connect the evidence (3 min) : Task: In your book: Watch the connection; name the limit. No writing yet. Staff move: Model the evidence-to-reason connection; keep the stated limitation visible.

We do · Phase or eclipse? (5 min) : Task: In your book: Record one placement and its science reason. Staff move: Pupil operates or directs. Name back the clue before any correction. Use the lightest prompt and fade it.

You do · explain and improve (9 min) : Task: In your book: Record evidence, explain it, then improve one part. Staff move: One adult supports the pupils needing access; the other holds the group, then swap roles across the week. Keep the first answer; give VF and one NS+; pupil edits // in the same colour on the same page.

Review and improve (3 min) : Task: In your book: Review one explanation; make one edit and mark it //. Staff move: Use three minutes: hear one precise VF/NS+ (one minute), make a pupil-owned edit (one minute), then receive the explanation of the change (one minute). Keep the original meaning, use // and add R only after genuine adult audience.

Exit ticket (4 min) : Task: In your book: Answer the prompt; show one next step. Staff move: Read the response privately. Pupil voice plus adult audience closes R only when both happen. Use the response to choose the next teaching route.

Practical safety : Adult uses a low-power cool light; no lasers or looking at the Sun. No blackout: normal-light diagrams are equal routes. Keep the ball above the observer's head shadow so a false eclipse is not created. Agree before any light change; adult handles equipment and cutting. Keep movement clear.

Preparation and equipment : Cool LED torch, large foam ball, observer marker and four phase cards.

Planned adaptation — Cognition and learning : Arrival and independent: one decision at a time; show a worked example, provide the topic word bank and fade the scaffold when the reason is secure.

Planned adaptation — Communication and interaction : We do: private thinking and partner rehearsal; accept pointing, signing, drawing, speaking or exact scribing. The pupil directs the decision.

Planned adaptation — SEMH : Predictable route choice, quiet pause and small re-entry step. Receive uncertainty calmly and keep correction private.

Planned adaptation — Sensory and physical : Model and practical: large diagrams, stable equipment, seated or no-touch director route; follow the topic safety note.

Planned adaptation — EAL : Pre-teach the named science words with the model; permit first-language planning and rehearsal. EAL is not SEN and does not reduce the science goal.

EEF strategy line : Explicit instruction in I do; scaffolding in the independent frame; flexible grouping for rehearsal; metacognitive checking in review; technology or diagrams to remove access barriers.

Staff: confirm the planned adaptations actually used : Record which adaptation helped and why in the evaluation.

If behind — Must teach core : I can explain why the Moon appears to change shape as it orbits Earth.

If behind — Shorten or merge : Shorten repeated rehearsal after a secure hinge; retain the individual explanation, safety and exit.

If behind — Next lesson : Use this evidence to choose the next topic route.

If behind — Dependency : Pass on the checked explanation and unresolved misconception; a group product cannot replace individual evidence.

Staff evaluation — Date taught : Record the date taught.

Staff evaluation — Delivered as planned : Yes / Partly / No; explain the change.

Staff evaluation — Criteria met : All / Most / Some; name the criterion evidenced.

Staff evaluation — Adaptations made and why : Record the support used and its effect.

Staff evaluation — Change from the SoW and coverage impact : Name the content affected and the next teaching action.

Staff evaluation — Evidence banked : Record the book location or evidence labelled on the school's evidence app.

P18 · Individual target opportunities

Cognition and learning : Arrival and independent: one decision at a time; show a worked example, provide the topic word bank and fade the scaffold when the reason is secure. Use the pupil's own target and note the prompt level, not a diagnosis.

Communication and interaction : We do: private thinking and partner rehearsal; accept pointing, signing, drawing, speaking or exact scribing. The pupil directs the decision. Use the pupil's own target and note the prompt level, not a diagnosis.

SEMH : Predictable route choice, quiet pause and small re-entry step. Receive uncertainty calmly and keep correction private. Use the pupil's own target and note the prompt level, not a diagnosis.

Sensory and physical : Model and practical: large diagrams, stable equipment, seated or no-touch director route; follow the topic safety note. Use the pupil's own target and note the prompt level, not a diagnosis.

EAL : Pre-teach the named science words with the model; permit first-language planning and rehearsal. EAL is not SEN and does not reduce the science goal. Use the pupil's own target and note the prompt level, not a diagnosis.

Individual plans : Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.

First lesson back · staff card

- Before the lesson, scan the books for five minutes. Look for genuine coverage and unfinished explanations; do not assume a missing page means the pupil was absent.
- Ask the cover teacher and TAs what was taught, what was practised and what still needs teaching.

- Check the roll against the books so that absence and missing evidence are kept separate.
- Existing NS+ is history. Begin fresh feedback from what the pupil shows now; do not retrospectively close an old loop.
- After the check, use one read, once, to shape Autumn 2. Choose the next teaching route from the pupil's evidence; do not add another record.

Preparation and practical safety

Cool LED torch, large foam ball, observer marker and four phase cards.

Adult uses a low-power cool light; no lasers or looking at the Sun. No blackout: normal-light diagrams are equal routes. Keep the ball above the observer's head shadow so a false eclipse is not created. Agree before any light change; adult handles equipment and cutting. Keep movement clear.

Adult uses a low-power cool light; no lasers or looking at the Sun. No blackout: normal-light diagrams are equal routes. Keep the ball above the observer's head shadow so a false eclipse is not created. Agree before any light change; adult handles equipment and cutting. Keep movement clear.

Staff cut the activity pieces before the lesson. Move pieces for the task; pupils record the answer and reason in their books.

Illustrative data are reasoning examples only, never a record that a pupil completed the practical. Use actual observations for completion evidence.

Adult deployment and feedback

During the independent task, one adult works with supported pupils and the other holds the group; swap the role across the week. Use least-prompt-first. Pupil meaning stays separate from adult support.

VF: verbal feedback. WS: worked with support. I: independent. NS+: one specific next step. E: evidence on the school's evidence app. //: pupil self-edit. ?: read this back. R: pupil response and genuine adult audience both happened.

Voice + Audience = R. The pupil owns the response; the adult reads or hears it and names the change. No extra receipt, signature or retrospective closure. Keep the edit in the same colour on the same page.

Hinge routes

Secure: transfer. Mixed: compare two choices. Misconception: model a counterexample, then ask a fresh question. Access: change the response mode. Method or data concern: restore the control or locator before judging the claim.

Answers and teaching cues

Hinge

Our changing view of the sunlit half causes ordinary phases. Earth's shadow matters during a lunar eclipse.

Model steps and limits

- Torch is the Sun, ball is the Moon, observer is Earth. Keep the sunlight direction fixed.
- Half the Moon is illuminated by the Sun. As it orbits, our view of the illuminated half changes.
- When the visible lit half faces us, we see full Moon. When it faces away, we see new Moon.
- Trace the light path: Sun → Moon → observer on Earth. Name the Sun as light source and the Moon as reflecting object.
- The Moon orbits Earth in about 27.3 days, roughly a month. The phase cycle is about 29.5 days.
- Sketch the view beside each physical model position. Mark the sunlight direction.
- A lunar eclipse occurs when Earth blocks sunlight from reaching the Moon; the tilted orbit means this is not monthly.
- The model does not show realistic sizes, distances or the tilt of the Moon's orbit.

Activity answers

- Crescent: A crescent is our view of part of the sunlit half.
- Full Moon: Full Moon is an ordinary phase; it does not require Earth's shadow.
- New Moon: At new Moon the visible side is largely unlit.
- Dark red Moon: A lunar eclipse involves Earth's shadow.
- Own light: Moonlight is reflected sunlight.
- Every month: The Moon's tilted orbit prevents an eclipse every month.

Misconception repair

A scientific claim needs relevant evidence, a reason and a fair limit.

Moon statement answers

- The Moon reflects sunlight. True: the light reaching us is reflected sunlight.
- The Moon glows by itself. False: it reflects sunlight rather than making its own visible light.
- Ordinary phases are Earth's shadow. False: phases come from our changing view of the lit half; Earth's shadow matters during a lunar eclipse.
- The Moon orbits Earth in about a month. True: about 27.3 days for one orbit; the phase cycle is about 29.5 days.
- The Moon is only visible at night. False: the Moon can also be visible in daylight.

Arrival answer cues

Supported

- 1. Full Moon: the visible disc is illuminated.
- 2. Planets orbit the Sun.
- 3. Sunlight reflected from the Moon.
- 4. Point to the visible part of the illuminated half.

Independent

- 1. Useful: motion around a centre. Limit: scale or hand-driven motion.
- 2. Gravity.
- 3. Sunlight reaches the Moon and reflects towards the observer.
- 4. The Moon orbits Earth, changing our view of its illuminated half.

Aspire

- 1. An adult moving a model supplies the motion; it is not a measurement of gravity.
- 2. Orbit is travel around another body; rotation is turning on an axis.
- 3. Sunlight illuminates the Moon; reflected light travels towards the observer.
- 4. A phase changes our view of the lit half; a lunar eclipse blocks sunlight with Earth.

Exit and independent answers

Check the pupil's own evidence against the learning criteria. State that the Moon orbits Earth in roughly a month.; Explain that moonlight is reflected sunlight.; Explain phases using our changing view of the lit half. Retain first and revised versions. A claim must name relevant evidence and a scientific reason; keep the stated model limit. Supplied examples are not personal practical completion.

Further topic teaching prompts by stage

Lesson overview

- last lesson — Describe the movement of the Moon around the Earth.
- I can state that the Moon orbits Earth in roughly a month.
- I can explain that moonlight is reflected sunlight.
- I can explain that phases come from our changing view of the lit half.

Arrival · prior lesson retrieval

- Point to the part of your model that was accurate.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — Last lesson you built a Solar System model. Name one thing it showed well and one it could not show.
- Point to Earth in the picture.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — the topic: what holds the planets on their orbits around the Sun?
- Say the one thing we change in a fair test.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — the topic: why do we repeat measurements?
- Look at the Moon pictures. Point to the fullest one.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — The Moon looks a different shape each week. Does the Moon itself really change?

- Last lesson you built a Solar System model. Name one thing it showed well and one it could not show.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — Last lesson: explain how you decided what your model was allowed to get wrong.
- the topic: what holds the planets on their orbits around the Sun?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — the topic: explain how movement and gravity together produce an orbit.
- the topic: why do we repeat measurements?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — the topic: explain how you would check an observation you could not repeat.
- The Moon looks a different shape each week. Does the Moon itself really change?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — The Sun always lights half the Moon. So why do we see different shapes?
- Last lesson: explain how you decided what your model was allowed to get wrong.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Point to the part of your model that was accurate.
- the topic: explain how movement and gravity together produce an orbit.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Point to Earth in the picture.
- the topic: explain how you would check an observation you could not repeat.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Say the one thing we change in a fair test.
- The Sun always lights half the Moon. So why do we see different shapes?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Look at the Moon pictures. Point to the fullest one.

Starter · make the new idea worth knowing

- Look at four Moon pictures: crescent, half, full, almost invisible. What might be changing? You do not need the answer yet — choose a hypothesis.
- Before we look: does the Moon itself change shape?
- Choose one. Say it, point to it, or press it.
- Nothing here is right or wrong yet. Nothing is marked, scored or stored. A prediction still counts when it turns out differently. That is what makes the test worth running.

I Do · model the first concept

- ► Reveal next thinking step

We Do · hinge question

- The Moon switching its own light on and off

We Do 2 · interactive exploration

- Choose a phase. Connect Moon position, sunlight direction and the view from Earth.

⇒ Independent task · evidence of thinking

- ⇒ Your independent product / decision
- Make or sequence a four-position phase strip or wheel. Do this for each position: SUNLIGHT DIRECTION → LIT HALF → WHAT WE SEE FROM EARTH. Then write one sentence about eclipses. Say how a phase and an eclipse differ.
- A photo on its own is not proof. Add what the pupil decided, explained, found or judged.
- Use the scientific word first. Ask for help only when you need it. TA fade route: model the word. Then prompt for it. Then wait for it. Drop a step as soon as the pupil no longer needs it.

Exit ticket · response becomes the next step

- Does the Moon make its own visible light?
- Explain why the Moon appears to change shape during the month.
- Explain why normal phases are not caused by Earth's shadow.
- 1) Pupil answers the Science question. 2) Adult genuinely receives it and gives one next step. 3) Pupil writes: "Next I will _____."
- No adult signature, initial or second receipt mark. If nothing needs changing, the next step can be transfer to a new example or greater independence.

-  GROW written close — the pupil writes this line
- “What I said, and what it changed.”
- No adult initial is required. The adult is audience for this line, not verifier. There is no signature, initial, tick or receipt mark. The line closes when it is written and received.
- If writing is hard: say it, and an adult writes down your exact words. Or point to what changed and say it aloud. Or draw it and give it one spoken caption. Or record it in the school's own workflow. The line always stays. The route changes how you answer, not the Science.

Lesson overview

- Moon Phases: Model and Explain
- I can use the model safely.
- I can identify new, quarter and full positions.
- I can explain what the model shows and how an eclipse is different.

Arrival · prior lesson retrieval

- Point to the Full Moon picture.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — Last lesson you explained why the Moon seems to change shape. What is really changing?
- Point to the Sun in the model.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — the topic: what keeps the planets travelling around the Sun?
- Point to something that turns.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — the topic: what does a mechanism change about a force?
- We make the phases with a ball and a lamp. Point to the lamp.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Independent — Today you make the phases appear on command. What will the lamp stand for?
- Last lesson you explained why the Moon seems to change shape. What is really changing?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — Last lesson: explain why the lit half of the Moon stays the same even though our view does not.
- the topic: what keeps the planets travelling around the Sun?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — the topic: explain why an orbit needs motion and gravity acting together.
- the topic: what does a mechanism change about a force?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — the topic: explain what a model of a mechanism can show and what it hides.
- Today you make the phases appear on command. What will the lamp stand for?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Aspire — A phase and an eclipse look related but are not the same. What must your model separate?
- Last lesson: explain why the lit half of the Moon stays the same even though our view does not.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Point to the Full Moon picture.
- the topic: explain why an orbit needs motion and gravity acting together.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Point to the Sun in the model.
- the topic: explain what a model of a mechanism can show and what it hides.
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — Point to something that turns.
- A phase and an eclipse look related but are not the same. What must your model separate?
- Non-reading route: point to it, say it, or show it. An adult can write down your exact words. Next step up: Supported — We make the phases with a ball and a lamp. Point to the lamp.

Starter · make the new idea worth knowing

- Predict what the observer will see when the Moon is between Earth and the Sun in our model. Commit before the light/ball moves.
- Before we model it: what makes a phase?

- Earth's shadow on the Moon

We Do · hinge question

- It must happen every month

We Do 2 · interactive exploration

- The phase explorer above shows the phases. This lab is about telling them apart from eclipses, which look related and are not.
- ▲ Earth's shadow on the Moon
- ☉ Which lit part we can see
- Choose one. Say it, point to it, or press it. Nothing here is right or wrong yet.
- Use your frozen list as the evidence. Decide, then say why.
- Every step is a button, and each option carries a shape and a word as well as its colour. Say it, point to it or have an adult press it — the route does not change the Science.

Independent task · evidence of thinking

- Work in pairs or in a group set by your teacher. Run the torch-and-ball model. Stop at four positions. Sketch or choose what the observer sees. Then build or label a phase wheel. Mark the sunlight direction on it.

Exit ticket · response becomes the next step

- How long does the Moon take to orbit Earth, roughly?
- Use the model to explain one Moon phase.
- Explain why an eclipse is different from the normal phase cycle and name one model limitation.

Title

- I can state that the Moon orbits the Earth, roughly once a month.
- I can explain that the Moon does not make its own light.
- I can describe why the Moon appears to change shape.

Arrival Task

- Arrival task — pick your tier
- Point to the full moon on the phase strip.
- Write how long you think the Moon takes to go round the Earth.
- Write how long the Moon takes to orbit Earth, and where its light comes from.

Today at a Glance

- Where the light comes from

I Do 1

- Watch me make the phases happen
- Torch-and-ball model: sunlight lights one half of the Moon always; as it orbits, our angle changes, so we see NEW, HALF, FULL. The phases are not the Earth's shadow.
- satellite — An object that orbits a planet. The Moon is Earth's natural satellite.
- phase — The shape of the lit part of the Moon that we can see.
- reflect — To bounce light off a surface.
- crescent — A thin curved phase.

I Do 2

- Stated that the Moon orbits Earth in about a month.
- Explained that moonlight is reflected sunlight.
- Modelled or described the cause of the phases.
- Capture: Photograph the pupil's completed phase wheel with the sunlight direction marked.

Independent Work

- Your turn — pick your tier
- Arrange or draw new, half and full Moon examples with an adult. Direct one model position and point to the lit part you can see. Correct a matching label if needed.
- Build and label a phase wheel or three-view strip. Check three positions with the model and explain why the visible lit part changes although the Moon stays round.

- Check the three views and explain the phases. Distinguish a lunar eclipse from the regular phase cycle, and explain one limitation of the torch/ball model.
- 4 min: Retrieve the last lesson's lit-side rule. Choose a pre-cut phase wheel, simple drawings or directing an adult's model; agree before any light level changes.
- 20 min: Build or arrange new, half and full Moon views. Use an adult-held torch and ball at three positions, keeping the light direction fixed and the ball slightly above head-shadow height. Compare the visible lit portion with the wheel.
- 2 min: Pens down if you are writing; pause and check. Ask an adult or chosen partner to check one view from the same observer position. Did the ball change shape, or did the visible lit part change?
- 6 min: Correct one phase label or view, explain what the model showed, and reset the room lighting and kit. A still-diagram explanation is an equal route.
- Access and equipment: Adult-prepared card circles, crayons, a ball and a low-power torch. Never point a torch into eyes; no looking at the Sun, lasers or blackout. Keep ordinary room light or use the supplied still model if dimming is unwelcome. An adult can operate all equipment.
- Drawing route: draw three equal circles; shade the part not seen lit for new, half and full. The circle itself stays round.
- Times to distinguish: one orbit relative to the stars takes about 27.3 days; the new-to-new phase cycle takes about 29.5 days. Both are roughly a month.
- Original classroom diagrams and tasks; the sources below support the facts. No external page is needed to do the activity.
- NASA: Moon phases — <https://science.nasa.gov/moon/moon-phases/>
- NASA Space Place: Moon phases and orbital periods — <https://spaceplace.nasa.gov/moon-phases/>

Lundy Loop

- Where what you say changes something
- The torch model needs low light. You will get a warning before the lights change, every time.
- Point, sign, say or show one observation, question or access need. You may pass or speak privately.
- Tell the adult teaching this lesson, or ask a chosen partner to help you pass on the message.
- The adult says what can change now, makes one agreed change if possible, and explains any limit. Check whether the change helped; no display or sharing is assumed.

Exit Ticket

- Does the Moon make its own light? Yes / No
- How long does the Moon take to orbit the Earth?
- About 27.3 days for one orbit, roughly a month. The phase cycle is about 29.5 days.
- Explain why phases are not caused by the Earth's shadow.
- Phases happen every month and depend on viewing angle; Earth's shadow only reaches the Moon during a rare eclipse.

Arrival and Starter

1. Name one useful feature and one limit of an orbit model.
2. What force keeps planets in orbit?
3. Explain where moonlight comes from.
4. Predict why the visible lit part changes.

Starter: Look at new, quarter and full Moon. What might be changing?

Method and safe participation

1. Keep the cool light fixed and use one Moon model or the prepared phase cards.
2. Mark the lit half and the observer position. Stop and check the light direction with the adult.
3. Sequence four views from the same observer, keeping phase and eclipse explanations separate.
4. In your book, label an observation and conclude why the visible lit part changes. Restore lighting and return equipment.

Only the apparatus for the current stage is on the table. If a pupil cannot safely take part, use a teacher demonstration and the same observation-and-conclusion task in the book.